SOLUTION

SECTION–I

1. Ans. (D)

$S_1P + S_2P = 2a$ & $S_1P - S_2P = 2A$

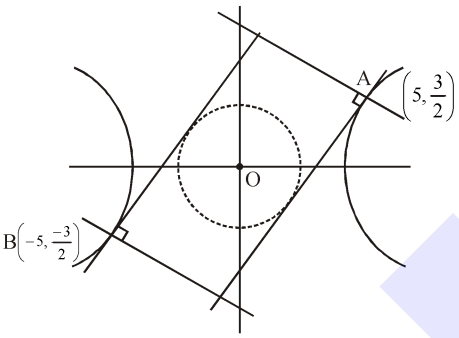
$S_1P = a + A$ & $S_2P = (a - A)$

quadratic equation is $x^2 - 2ax + (a^2 - A^2) = 0$

2. Ans. (D)

Sol. $\frac{x^2}{16} - \frac{y^2}{4} = 1$

$\left(5, \frac{3}{2}\right)$ lies on the curve.



tangent at A

$\frac{5x}{16} - \frac{3}{2}\left(\frac{y}{4}\right) = 1 \Rightarrow 5x - 6y = 16$

tangent at B

$5x - 6y = -6$

Normal at A

$6x + 5y = l$

A $\left(5, \frac{3}{2}\right)$ $30 + \frac{15}{2} = l$

$\Rightarrow \lambda = \frac{75}{2}$ $6x + 5y = \frac{75}{2}$

$|||^{ly}$ normal at B is $6x + 5y = \frac{-75}{2}$ distance between tangents is $2r = \frac{32}{\sqrt{25+36}}$

$\therefore r = \frac{16}{\sqrt{61}}$

Radius of circumcircle of rectangle is

$R = \sqrt{25 + \frac{9}{4}} = \frac{\sqrt{109}}{2}$

3. Ans. (B)

Let the point be $(\alpha,\beta) \Rightarrow \beta = \alpha + 2$

Chord of contact of hyperbola is $T = 0$

$\Rightarrow \frac{\alpha x}{6} - \frac{\beta y}{2} = 1 \Rightarrow \frac{\alpha x}{6} - \frac{(\alpha+2)y}{2} = 1$

$\Rightarrow \alpha\left(\frac{x}{6} - \frac{y}{2}\right) - \left(\frac{2y}{2} + 1\right) = 0$

Since this passes through (λ,μ)

$\therefore \frac{\lambda}{6} - \frac{\mu}{2} = 0$ and $\frac{2\mu}{2} + 1 = 0 \Rightarrow \frac{\lambda}{\mu} = 3$

4. Ans. (D)

The chord of contact of (2, 3) is $3y = 2(x + 2)$

solving it with $y^2 = 4x$

$y^2 = 2(3y - 4) \Rightarrow y^2 - 6y + 8 = 0 \Rightarrow y = 2, 4$

so $x = 1$ & 4 respectively so points are (1, 2) and (4, 4).

5. Ans. (A)

Equation of the normal to the parabola $y^2 = 12x$ is given by

$y = mx - 6m - 3m^3$

It passes through (1, 2) so

$2 = m - 6m - 3m^3$

$\Rightarrow 3m^3 + 5m + 2 = 0$

$m_1m_2 + m_2m_3 + m_3m_1 = 5/3$ ...(i)

$m_1m_2m_3 = -2/3$ ...(ii)

(i) divided by (ii)

$\frac{1}{m_1} + \frac{1}{m_2} + \frac{1}{m_3} = \frac{-5}{2}$.

6. Ans. (C)

Any tangent to $y = x^2$ can be written as

$y = mx - \frac{1}{4}m^2$ if it touches 2nd, then

$mx - \frac{m^2}{4} = x^2 - 2x + 2$

$\Rightarrow x^2 - (2+m)x + \left(2 + \frac{m^2}{4}\right) = 0$

$D = 0 \Rightarrow (m+2)^2 - 4\left(2 + \frac{m^2}{4}\right)$

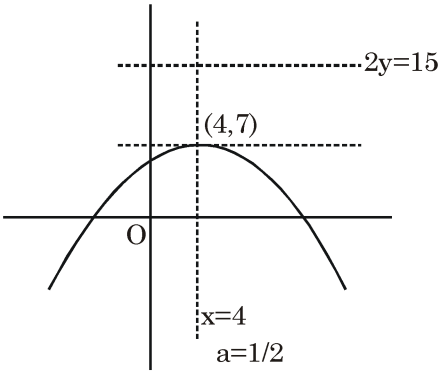
$4m - 4 = 0 \Rightarrow m = 1 \Rightarrow 4x - 4y - 1 = 0$

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7. Ans. (A)

$x^2 - 2.4x + 16 = 16 - 2 - 2y$

$(x-4)^2 = -2(y-7)$



8. Ans. (B)

Focus $\equiv (2, 0)$, point $\equiv (2t^2, 4t)$

$\Rightarrow \quad h = \dfrac{2+2t^2}{2}$

$k = \dfrac{4t+0}{2}$

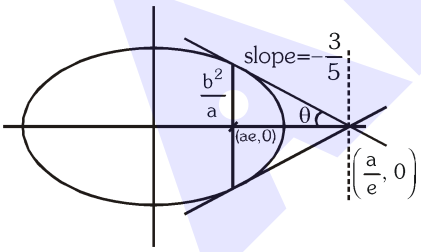
$\Rightarrow \quad h = 1 + t^2 \;\&\; k = 2t$

$\Rightarrow \quad h = 1 + \dfrac{k^2}{4} \quad \Rightarrow \quad y^2 = 4x - 4$

9. Ans. (D)

Point of intersection of tangents is $\left(\dfrac{20}{3}, 0\right)$ which lie on the corresponding directrix of the ellipse

$\therefore \quad \dfrac{a}{e} = \dfrac{20}{3}$



$\tan\theta = \dfrac{b^2}{a\left(\frac{a}{e} - ae\right)} = \dfrac{b^2 e}{a^2(1-e^2)} = e$

$\dfrac{3}{5} = e \qquad \therefore \quad a = \dfrac{20}{3} \times \dfrac{3}{5} = 4$

$b^2 = 16\left(1 - \dfrac{9}{25}\right) = \dfrac{16 \times 16}{25}$

$\ell_{LR} = \dfrac{2b^2}{a} = \dfrac{2 \times 256}{4 \times 25} = 5.12$

10. Ans. (D)

The vertices of the square will lie on director's circle of ellipse $\dfrac{x^2}{16} + \dfrac{y^2}{b^2} = 1$

$\Rightarrow$ vertices of square lies on $x^2 + y^2 = 16 + b^2$

$\therefore$ $2(\text{side of square})^2 = (\text{diameter of director's circle})^2$

$\Rightarrow$ Radius of director's circle = 5

$\Rightarrow \quad 16 + b^2 = 25 \;\text{ or }\; b = \pm 3$

$\therefore$ Area of ellipse = $\pi(4)(3)$ sq. units

$= 12\pi$ sq. units

11. Ans. (B,D)

Let the common tangent be

$y = mx \pm \sqrt{12m^2 - 12}$ solving it with

$xy = 8 \quad \Rightarrow \quad mx^2 \pm \sqrt{12(m^2-1)}x - 8 = 0$

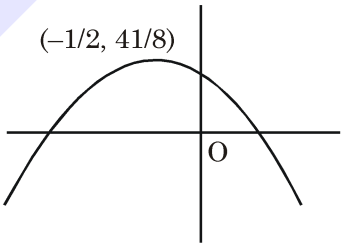
$D = 0 \quad \Rightarrow \quad 12(m^2 - 1) + 32m = 0$

$3m^2 + 8m - 3 = 0 \;\Rightarrow\; m = -3, \dfrac{1}{3}$

$y = -3x \pm 4\sqrt{6} \qquad m = \dfrac{1}{3}$ rejected

12. Ans. (A,D)

$(f(0) - 5)^2 + (f(1) - 4)^2 + (f(2) - 2)^2 = 0$

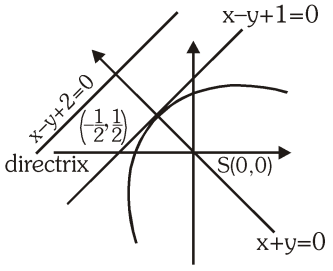


$f(0) = 5, \; f(1) = 4, \; f(2) = 2$

$f(x) = -\dfrac{x^2}{2} - \dfrac{x}{2} + 5$

13. Ans. (A,C)

$(x + y)^2 = 4(x - y + 1)$

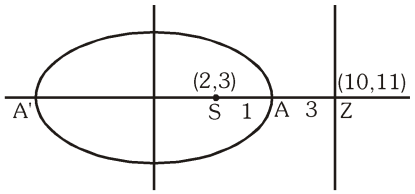


$2\left(\dfrac{x+y}{\sqrt{2}}\right)^2 = 4\sqrt{2}\left(\dfrac{x-y+1}{\sqrt{2}}\right)$

$\left(\dfrac{x+y}{\sqrt{2}}\right)^2 = 2\sqrt{2}\left(\dfrac{x-y+1}{\sqrt{2}}\right)$

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14. Ans. (A,B,C)



$A \equiv \left(\frac{10+6}{4}, \frac{11+9}{4}\right) = (4,5)$

$A' \equiv \left(\frac{10-6}{1-3}, \frac{11-9}{1-3}\right) = (-2,-1)$

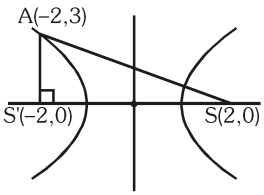
Major axis, AA' = $\sqrt{36+36} = 6\sqrt{2}$

Minor axis $2b = 2a\sqrt{1-e^2} = 2\left(3\sqrt{2}\right)\sqrt{1-\frac{1}{9}} = 8$

Latus rectum = $\frac{2b^2}{a} = \frac{2\times 16}{3\sqrt{2}} = \frac{16\sqrt{2}}{3}$

15. Ans. (A,B,C,D)

S ≡ (2, 0), S' ≡ (–2, 0)

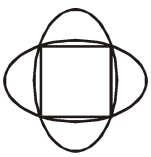


Using reflection property of hyperbola,
S'A is incident ray.
Equation of incident ray S'A is x = –2
Equation of reflected ray SP is 3x + 4y = 6.
Now 2ae = 4 ⇒ ae = 2 ....(i)
Point (–2, 3) lies on hyperbola,

$\therefore \quad \frac{4}{a^2} - \frac{9}{b^2} = 1 \Rightarrow \frac{4}{a^2} - \frac{9}{4-a^2} = 1$

on solving it we get a = 4 (reject), a = 1 .....(ii)
∴ Using (i) & (ii), we get e = 2
length of latus rectum = $2a(e^2 - 1) = 6$

16. Ans. (A,C)



The given points lie on

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ & $\frac{x^2}{b^2} + \frac{y^2}{a^2} = 1$

The given points lie on a square & no parabola can pass through all vertices of a square

Paragraph for Question 17 & 18

17. Ans. (B)

Minimum value occurs at x = 2
so $f(2) = |a-2| = 1 \Rightarrow a = 3, 1$
$\Rightarrow$ a = 3 (as > 2)

$\Rightarrow$ Ellipse is $\frac{x^2}{9} + \frac{y^2}{4} = 1$

& $e = \sqrt{1-\frac{4}{9}} = \frac{\sqrt{5}}{3}$

18. Ans. (D)

Let tangent AB is $\frac{x\cos\theta}{3} + \frac{y\sin\theta}{2} = 1$
AB passes through P(1, 2)

$\Rightarrow \quad \frac{\cos\theta}{3} + \sin\theta = 1$

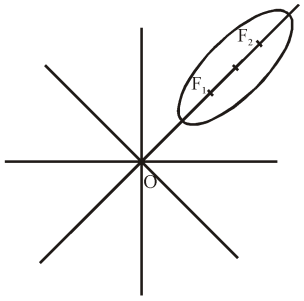
$\theta = \frac{\pi}{2}$ is one solution

also $\frac{1}{3} + \tan\theta = \sec\theta \Rightarrow \sec\theta - \tan\theta = \frac{1}{3}$ ....(i)
& $\sec\theta + \tan\theta = 3$ .....(ii)
$\Rightarrow \quad \sec\theta = \frac{5}{3}$ & $\tan\theta = \frac{4}{3}$
Hence none of the options necessarily are true, so (D)

Paragraph for Question 19 & 20

19. Ans. (D)

Focus of C_2 is $(\sqrt{2}, \sqrt{2})$ &
asymptote of C_1 is y = –x & y = x.
∴ Focus of C_3 is $F_1(\sqrt{2}, \sqrt{2})$ directrix is y = –x
for curve C_3



$a\left(\frac{1}{e} - e\right) = 2$

$\frac{4}{3}\left(\frac{1}{e} - e\right) = 2$

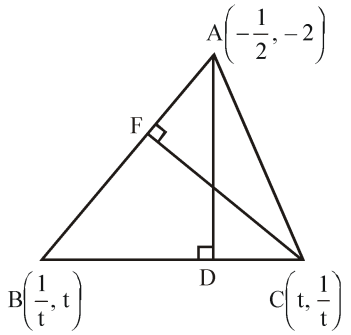
$\Rightarrow \quad e = \frac{1}{2}$

20. Ans. (A)

$A\left(-\frac{1}{2}, -2\right)$

Let $B\left(\frac{1}{t}, t\right)$,

then $C\left(t, \frac{1}{t}\right)$



Slope of BC = –1 ⇒ Slope of AD = 1

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Equation of AD : $y + 2 = x + \frac{1}{2}$

or $2x - 2y = 3$ .....(1)

Now slope of AB = 2t

$\Rightarrow$ Slope of $CF = -\frac{1}{2t}$

$\Rightarrow$ equation of CF : $y - \frac{1}{t} = -\frac{1}{2t}(x - t)$

or $x + 2ty = t + 2$ .....(2)

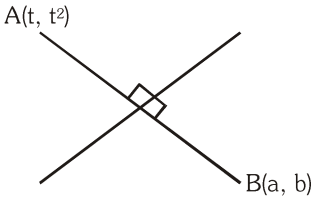
Solving (1) & (2), we get

$\therefore$ orthocentre is $\left(2, \frac{1}{2}\right)$

SECTION–III

1. Ans. 2

General point on $y = x^2$ in (t, t^2) let us find the image of (t, t^2) in $x + y = 1$



$$\frac{a+t}{2} + \frac{b+t^2}{2} = 1$$

$$a + t + b + t^2 = 2 \quad ...(1)$$

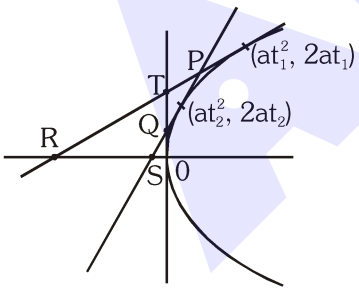
and $\frac{b-t^2}{a-t} = 1$ ...(2)

eliminate t from (1) & (2)

$a + b^2 = 2b$

$\Rightarrow x + y^2 = 2y \Rightarrow k = 2$

2. Ans. 1



The equation of first tangent is $t_1y = x + at_1^2$

$T(0, at_1)$ & $R(-at_1^2, 0)$

For second tangent

$Q(0, at_2)$ & $S(-at_2^2, 0)$

For concyclic points (OT) (OQ) = (OR) (OS)

$\Rightarrow t_1t_2 = 1$

Point $x = at_1t_2$

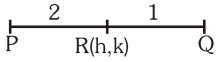
$\Rightarrow x = a$

$\Rightarrow \lambda_1 = 1$

3. Ans. 9

Equation of tangent is $\frac{x\cos\theta}{a} + \frac{y\sin\theta}{b} = 1$

$P\left(\frac{a}{\cos\theta}, 0\right)$ & $Q\left(0, \frac{b}{\sin\theta}\right)$



$h = \frac{a}{3\cos\theta}$ & $k = \frac{2b}{3\sin\theta}$

$\cos\theta = \frac{a}{3h}$ & $\sin\theta = \frac{2b}{3k}$

$\frac{a^2}{h^2} + \frac{4b^2}{k^2} = 9 \Rightarrow \frac{a^2}{x^2} + \frac{4b^2}{y^2} = 9$

4. Ans. 1

A tangent to $\frac{x^2}{16} - \frac{y^2}{9} = 1$ is

$$y = m_1x \pm \sqrt{16m_1^2 - 9} \quad .....(i)$$

The other hyperbola is $\frac{x^2}{-9} - \frac{y^2}{-16} = 1$

then tangent to it is

$y = m_2x \pm \sqrt{-9m_2^2 - (-16)}$ .....(ii)

(i) & (ii) are same.

$\Rightarrow m_1 = m_2$ & $16m_1^2 - 9 = -9m_2^2 + 16$

$\Rightarrow 25m_1^2 = 25 \Rightarrow m_1^2 = 1 \Rightarrow m_1 = \pm 1$

$\therefore m = 1$

5. Ans. 5

Assume (h,k) be the middle point of chord

$\therefore$ Its equation is

$hx - ky = h^2 - k^2 \Rightarrow y = \frac{h}{k}x + \frac{k^2 - h^2}{k}$

Now using $c = \frac{a}{m}$, we get

$\frac{k^2 - h^2}{k} = \frac{5}{h}.k$

$\Rightarrow x^3 = y^2(x - 5)$

$\therefore a = 5$

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